

A Simple Property of the Digits of Multi-digit Primes

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Abstract

We present and prove a curious property of prime numbers written in base 10: if a prime has at least two digits, then the greatest common divisor (gcd) of its decimal digits must be equal to 1. This property gives a simple necessary condition for primality and highlights an interplay between number-theoretic divisibility and digit structure.

1 Introduction

Prime numbers are fundamental in mathematics. Many characterizations of primes involve modular arithmetic, such as Wilson's theorem or Fermat's little theorem. In this short note, we present a very elementary but lesser-known property relating to the decimal digits of primes.

2 Main Result

Theorem 1 (Digit Coprimality of Multi-digit Primes). *Let p be a prime with at least two decimal digits. Then the greatest common divisor of the decimal digits of p is equal to 1.*

Proof. Write the decimal expansion of p as

$$p = \sum_{i=0}^r a_i 10^i,$$

where $a_0, a_1, \dots, a_r$ are the digits of p with $a_r \neq 0$. Let

$$d = \gcd(a_0, a_1, \dots, a_r).$$

Suppose, for contradiction, that $d > 1$. Then each a_i is divisible by d , say $a_i = db_i$. Substituting, we obtain

$$p = \sum_{i=0}^r (db_i) 10^i = d \cdot \left(\sum_{i=0}^r b_i 10^i \right).$$

Thus $d \mid p$.

Since $1 < d \leq 9$, there are two possibilities:

- If $d = p$, then $p \leq 9$. This contradicts the assumption that p has at least two digits.
- If $1 < d < p$, then p is divisible by d and therefore composite, contradicting primality.

In both cases, we obtain a contradiction. Hence $d = 1$. □

3 Examples

Example 1. $p = 13$: digits are 1 and 3, with $\gcd(1, 3) = 1$. Prime.

Example 2. $p = 37$: digits are 3 and 7, with $\gcd(3, 7) = 1$. Prime.

Example 3. $p = 119$: digits are 1, 1, 9, with $\gcd(1, 1, 9) = 1$, but $119 = 7 \times 17$. This illustrates that the theorem only gives a necessary condition, not a sufficient one.

4 Remarks

Remark 1. The theorem excludes single-digit primes 2, 3, 5, 7, for which the gcd of the digits equals the number itself. Thus the condition “at least two digits” is essential.

Remark 2. The result provides a simple filter to rule out some composite numbers: if the digits of a number have $\gcd > 1$, the number cannot be prime.

5 Conclusion

We have shown that for multi-digit primes, the gcd of the digits is always 1. This simple property illustrates an unexpected link between base-10 representation and primality. Although not a primality test, it serves as a useful necessary condition and a charming curiosity in number theory.